25900016 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCBTS100 - BIOLOGICAL FOUNDATIONS FOR HEALTH AND WELLNESS

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What is the term for the different forms of a single gene? [K] / [CO3]

   a). Allele

   b). Locus

   c). Genotype

   d). Chromosomes

2. What is the function of the ovary in a flowering plant? [U] / [CO3]

   a). To produce pollen

   b). To produce ovules

   c). To attract polinators

   d). To hold anthers

3. What is the primary female reproductive hormone? [K] / [CO2]

a). Estrogen

b). Testosterone

c). insulin

d). adrenaline

4. What is the main role of GABA in the brain? [U] / [CO5]

a). Excitatory signaling

b). Muscle contraction

c). Memory enhancement

d). Inhibitory neurotransmission

5. What connects muscles to bones? [K] / [CO2]

a). Ligaments

b). Tendons

c). Cartilage

d). Muscles

[1x5 = 5]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

6. Give two functions of carbohydrates. [U] / [CO1]

7. How would damage to the pituitary gland affect other endocrine glands? [A] / [CO5]

8. What role does protein play in muscle repair and growth? [U] / [CO4]

9. What are proteins? Give examples. [K] / [CO1]

10. What is the excretory system? [K] / [CO2]

11. Which level includes all the living and non living components in an area? [K] / [CO1]

12. Explain the difference between innate and adaptive immunity. [U] / [CO5]

[2x5 = 10]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

13. Explain the factors enhancing physical and mental health.	[K] / [CO1]
14. Explain the different chronic diseases due to junk food consumption.	[U] / [CO4]
15. Explain gene expression.	[K] / [CO3]
16. How does neurotransmitter imbalance contribute to mental health disorders like depression and anxiety?	[An] / [CO5]
17. Name the major organs of the digestive system and their functions.	[K] / [CO2]
18. What is the circulatory system, and why is it important?	[K] / [CO2]
19. What are preventive measures for maintaining a healthy respiratory system?	[K] / [CO2]

[4x5 = 20]